Mock Round 1 — 25 MCQ Questions

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Section 1 — Guess the Output from Code (5)

Q1

```
n = 4
for i in range(1, n + 1):
    print(i * "*")
```

- A) ⭐

  ```
  *
  **
  ***
  ****
  ```

- B)

  ```
  ****
  ***
  **
  *
  ```

- C)

  ```
  *
  *
  *
  *
  ```

- D)

  ```
  1
  22
  333
  4444
  ```

Q2

```
x = 5
while x > 0:
    print(x, end=" ")
    x -= 2
```

- A) ⭐ 5 3 1
- B) 5 4 3 2 1
- C) 5 3
- D) 1 3 5

Q3

```
s = "abcdefgh"
print(s[::-2])
```

- A) abc
- B) cba
- C) ⭐ hfdb
- D) Error

Q4

```
a = ['1', '2', '3']
print(a[-1] + a[0])
```

- A) 4
- B) 21
- C) ⭐ 31
- D) Error

Q5

```
def f(n):
    if n == 0:
        return
    print(n, end=" ")
    f(n - 1)

f(3)
```

- A) 1 2 3
- B) ⭐ 3 2 1

- C) 3 2 1 0
- D) Infinite recursion

Section 2 — Programming Logic Questions (5)

Q6

```
x = 0
for i in range(1, 4):
    x += i
print(x)
```

- A) 3
- B) ⭐ 6
- C) 7
- D) 10

Q7

```
s = 0
for i in range(5):
    if i & 1:
        s += i
print(s)
```

- A) ⭐ 4
- B) 9
- C) 10
- D) 6

Q8 Which loop prints `1 4 9 16`?

- A) ⭐ `for i in range(1,5): print(i*i, end=' ')`
- B) `for i in range(4): print(i**2, end=' ')`
- C) `for i in range(1,5): print(i, end=' ')`
- D) `for i in range(1,5): print(i+1, end=' ')`

Q9

```
for i in range(3):
    print(chr(65 + i), end='')
```

- A) ⭐ ABC

- B) 123
- C) XYZ
- D) Error

Q10

```
count = 0
while count < 8:
    count += 2
```

- A) 2
- B) 3
- C) ⭐ 4
- D) 5

Section 3 — Debugging (5)

Q11

```
arr = [10, 20, 30, 40]
for i in range(1, 5):
    print(arr[i])
```

- A) Prints all
- B) ⭐ IndexError
- C) SyntaxError
- D) Infinite loop

Q12

```
x = 2
if x = 5:
    print("Yes")
```

- A) Runtime error
- B) ⭐ Syntax error
- C) Logical error
- D) No issue

Q13

```
a = 1
b = 3
def add(a, b):
    c = a + b
print(c)
```

- A) Correct sum
- B) ⭐ NameError
- C) SyntaxError
- D) None

Q14

```
nums = [1, 2, 3]
print(nums[2])
```

- A) ⭐ No Error
- B) Syntax Error
- C) IndexError
- D) Runtime Error

Q15

```
def fact(n):
    if n == 1:
        return 1
    return n * fact(n)
print(fact(5))
```

- A) Wrong base case
- B) ⭐ Missing decrement
- C) Wrong return type
- D) Indentation

Section 4 — Guess Code from Output (5)

Q16 Output: `[0, 1, 4, 9, 16]`

- A) `for i in range(5): print(i*i)`
- B) `for i in range(5): print(i)`
- C) `for i in range(1,5): print(i*i)`
- D) ⭐ `print([i*i for i in range(5)])`

Q17 Output:

```
A
BB
CCC
```

- A) ⭐ `for i in range(3): print(chr(65+i)*(i+1))`
- B) `for i in range(3): print(chr(65+i))`
- C) `for i in range(1,4): print(i*i)`
- D) `print('ABC')`

Q18 Output: `87654321`

- A) ⭐ `for i in range(8,0,-1): print(i, end='')`
- B) `for i in range(1,9): print(i, end='')`
- C) `print('12345678')`
- D) `for i in range(8,1,-1): print(i, end='')`

Q19 Output:

```
1
01
101
0101
```

- A) ⭐ `for i in range(1,5): print(('01'*i)[-i:])`
- B) `for i in range(4): print('10')`
- C) `for i in range(4): print('1010'*i)`
- D) `for i in range(1,5): print(('01'*i)[:-i])`

Q20 Output:

```
1
22
333
4444
```

- A) ⭐ `for i in range(1,5): print(str(i)*i)`
- B) `for i in range(1,5): print(i)`
- C) `print(1234)`
- D) `for i in range(4): print(i*i)`

Section 5 — DSA Puzzle Hunt (Harder) (5)

Q21 Which condition correctly checks whether `n` is a power of 2?

- A) `n % 2 == 0`
- B) ⭐ `n & (n - 1) == 0`
- C) `n | (n - 1) == 0`
- D) `n ^ 2 == 0`

Q22 Which data structure works on FIFO?

- A) Stack
- B) ⭐ Queue
- C) Tree
- D) Graph

Q23 You are given a square matrix `N x N`. Which is the correct in-place logic to rotate the matrix 90° clockwise?

- A) ⭐ transpose(matrix) → reverse each row
- B) transpose(matrix) → reverse each column
- C) reverse each row → transpose(matrix)
- D) reverse each column → transpose(matrix)

Q24 A robot moves only right or down in a `3 x 3` grid. Unique paths?

- A) 4
- B) ⭐ 6
- C) 8
- D) 9

Q25 There are 100 light bulbs lined up in a row in a long room. Each bulb has its own switch and is currently switched off. The room has an entry door and an exit door. There are 100 people lined up outside the entry door. Each bulb is numbered consecutively from 1 to 100. So is each person. Person No. 1 enters the room, switches on every bulb, and exits. Person No. 2 enters and flips the switch on every second bulb. Person No. 3 enters and flips the switch on every third bulb. This continues until all 100 people have passed through the room. How many of the light bulbs are illuminated after the 100th person has passed through the room?

- A) 37
- B) 50

- C) ⭐ 10
- D) 15